

Surveillance Car with 360° Camera, Environmental Monitoring, and Obstacle Detection and Avoidance System(S.H.I.E.L.D.)

An IoT enabled mobile car with live video, environmental monitoring and smart obstacle avoidance for autonomous patrolling.

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Abstract—In this article, the process of designing and development, as well as the implementation, of an inexpensive remote-controlled mobile surveillance system based on Internet of Things technology is described. The aim in this consideration is to develop a robotic vehicle roaming in the assigned zone, transmitting real-time images, and possessing enough intelligence to act accordingly. The objectives are achieved by creating an architecture of two microcontrollers.

In this setup, the Arduino Uno is the center for sensing, which is required for gathering information on the surroundings and the proximity sensors, and the main task is to sense any nearby obstacles. The ESP32-CAM is the controlling center as regards communication and mobility, as it is the part of the system controlling the motion of the motor and the camera.

The two controllers will interact in this system as follows: if there is an obstacle detected by the Arduino system within the path of the vehicle, an instant alert will be produced for the ESP32-CAM system. The action mentioned will automatically stop the motor and hence no accident will happen, and simultaneously, an alert will be produced to the interface. The prototype mentioned here is linked with the real-time visual feedback and critical data from the sensors, and hence this is one of the most flexible and friendly systems to be used, for example, in neighborhood surveillance systems, robots for education, and even in the development of embedded systems.

Index Terms—Microcontroller, IoT, Sensor Interfacing, Embedded System, Obstacle Detection, Robotics, Environmental Monitoring.

I. INTRODUCTION

A. Background and Motivation

In this modern-day era, for efficient surveillance, whether it is for security and monitoring purposes, systems of complexity and flexibility are required, something which

was not the case in the old days. In today's era, innovative and interconnected technology is very much alive and in action, and yet, surprisingly, much of the current surveillance technology is very outdated, to say the least. Let's, for instance, think of the common security cameras. The presence of human patrols ensures monitoring is dynamic and flexible. The individuals are able to analyze situations, alter their path based on experience, and investigate any discrepancies. Notwithstanding, this monitoring technique is riddled with numerous demerits. The cost implications are very high, taking into consideration salaries and training on an on-going basis. More importantly, this monitoring technique may expose people to danger, either in industrial settings, which already have inherent risks, or simply in remote locations.

“What if we were able to build this mobile robotic sentry which combined the best of both worlds?” Meaning, we wanted to develop this low-cost platform on wheels, so we're able to go over this region and get this live, real Panhandle-type feed so we're not missing any spots. But we didn't want to just stop there. In other words, we were like, “Hey, we can add all this other stuff to this thing so it functions as our ‘senses,’ sensing all this other information, air quality, temperature, and all this other stuff.” But ultimately, our aim is to simply build this very useful guardian, this system which makes all this advanced monitoring more safe, more flexible, and more available to all customers, small business, large business, this type of thing.

B. Problem Statement

The most fundamental challenge in devising such a device is not just putting hardware together, but rather

integrating it all. Multiple complex functionalities into one single, reliable system. The main challenges we need to address are:

- 1) . Ensuring consistency and high quality within live video output from a moving platform is more difficult than it appears. The stream has to be low-latency and smooth any delay or choppiness that might prevent a remote operator from reacting to events in real time. The system needs to capture a video, compress it, use it effectively and transmit wirelessly, while the reason behind this is the potential differences in signal strength across varying environments.
- 2) Our robot is designed to be more than just mobile camera; it is also a roaming environmental station. It must collect data simultaneously and continuously from a diverse array of sensors—measuring temperature, hazardous gases detection, and quantification dust particles. Coordinating these various electronic and its components work together in harmony to ensure that data accuracy and timely processing are of utmost value.
- 3) For such a platform to be both intelligent and safe have a degree of autonomous navigation. Using ultrasonic sensors—similar to a bat’s echolocation—it needs constantly to scan its environment and make instant decisions are hence made to avoid obstacles like furniture and walls. This is advanced programming that can interpret sensor data in milliseconds, translate it, and turn it into accurate motor commands if collisions are to be avoided.

C. Project Objectives and Scope

Our Mission: To develop a functional robotic platform that is an expression of this integrated concept. For that purpose, The following specific objectives have been identified:

- 1) **Build the Body:** Build a strong, wheeled robotic body chassis capable of reliably carrying all the electronic components and maneuvering across typical indoor floor surfaces.
- 2) **Give It Eyes:** Use an ESP32-CAM module as the robot’s primary vision system, programmed to broadcast a real-time video stream that is accessible to remote users via a simple, intuitive web interface.
- 3) **Give It Senses:** Include a wide array of environmental sensors to build a complete picture of the environment around the robot is-
 - A temperature and humidity sensor to operate as a digital thermometer.
 - A gas sensor for the detection of unsafe air conditions.
 - A dust sensor for measuring particulate matter concentrations.
- 4) **Give It Reflexes:** Put ultrasonic sensors on the robot’s front and develop a control “brain”. That allows the device to automatically stop and evade around obstacles and avoiding accidents by reactive decisions.
- 5) **Create a Command Center:** Develop and implement a clean, user-friendly web-based dashboard to serve as a mission control. The interface will enable

users to View the live robot camera feed; observe real-time sensor data in clear graphs, and remotely control the movements of the robot using a virtual joystick.

The scope of the project focuses on the development of a proof- proof-of-concept that works well under normal indoor conditions, such as in homes or offices, the underlying Principles are scalable to larger or outdoor applications; the primary goal is the delivery of a robust and functional prototype that should validate the core thesis.

D. Related Works

Most previous work in this respect has relied on sensor integration for data collection, such as in the use of ultrasonic systems for traffic monitoring described in [1]. Such schemes perform well within their narrow application niches but often lack mobility and dynamic environment handling capability. Other works have used mobile robots; these are also mostly narrowly designed to perform just one function, such as air quality monitoring or video capture [2]. By contrast, our project stands out for its comprehensiveness and multi-functionality. Instead of confining the robot to one function, we have designed a versatile “multi-tool on wheels” that integrates into an easily deployable, affordable solution with the functionalities of a portable surveillance camera, an environmental sensor suite, and a semi-independent navigation system.

Powered by an economical yet robust microcontroller with an integrated camera, such as the ESP32, our system demonstrates that an advanced monitoring device is capable of visual observation, environmental sensing, and autonomous movement—all these without requiring substantial financial investment—to deliver thorough situational insights.

II. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture

Architecture of the System The system uses a master-slave dual-controller architecture to Distribute the processing load to improve real-time performance:

- 1) **Arduino Uno R3 (Sensor Manager):** It acts as the Dedicated Data Acquisition Unit. It is responsible for reading and displaying data from all environmental sensors (DHT11, MQ, Dust) and the Ultrasonic sensor on the 16x2 LCD. It transmits this processed sensor data to the ESP32-CAM via serial communication.
- 2) **ESP32-CAM (IoT and Command Hub):** Functions as the Master Controller. Its roles include:
 - a) Networking: Sets up the Wi-Fi Access Point and operates the internal web server.
 - b) Video Streaming: Manages the camera module to capture and stream live video in real-time.
 - c) Motion Control: Directly commands the L298N motor driver for vehicle movement and the servo for camera positioning.
 - d) Alert System: Interprets sensor data from the Arduino to execute an immediate stop and notify the user of nearby obstacles.

B. Hardware and Software Components

- Controllers: Arduino Uno R3, ESP32-CAM
- Sensors: DHT11, MQ Gas Sensor, Dust Sensor, Ultrasonic Sensor (HC-SR04)
- Actuators: Gear Motors, L298N Motor Driver, SG90 Servo Motor
- Power: Rechargeable Li-Ion Battery Pack
- Display: 16x2 LCD
- Software: Arduino IDE (C/C++), ESP32-CAM Web-Server libraries

C. Working Principle and Data Flow

- 1) Initialization: The ESP32-CAM initializes the network; the Arduino performs a calibration phase with a set of sensors.
- 2) Sensor and communication loop: The Arduino works as a data hub, which continuously gathers sensor readings and forwards them to the ESP32-CAM.
- 3) Command and Control: The ESP32-CAM acts as the command center; it reads user input from the web interface for control over vehicle motion and camera orientation.
- 4) Detection and Alert: The obstacle detection system generates an interrupt-like event. In case of a high-priority proximity warning, ESP32-CAM gives an immediate stop command and alerts the user.
- 5) Streaming: ESP32-CAM multiplexes the live video feed and updated sensor data in real time to the remote user interface.

III. IMPLEMENTATION AND RESULTS

A. System Setup

The prototype was built on an efficient 4-wheeled The 4WD chassis gave a stable platform for mounting the sensors as well as conducting mobility experiments. The full complement of sensing elements required, including Temperature and humidity sensing unit DHT11 gas sensing unit(MQ series gas sensor), dust sensor and Ultrasonic sensor mounted appropriately on the chassis. This would enable one to monitor the environment clearly. The Arduino Uno R3 and 16x2 LCD display were fixed on to Upper part of the chassis to enable easy monitoring of the sensed data. Higher level control parts in the same period, including the ESP32-CAM module, L298N motor driver, and the rechargeable battery pack was properly seated. on the chassis body. These are located so as to achieve appropriate center of gravity, reduce the effects of vibration Place it on the ESP32-CAM module and ensure good airflow around it. for the L298N motor driver. Besides this positioning, ensures that low-power sensing and high-power control The signals are well separated and distinct, hence reducing electrical noises and enhancing system robustness. The overall The hardware setting on the 4WD chassis ensures efficient and: It guarantees a stable motion with optimized integration of sensing. and control elements

B. Testing and Performance Outcomes

To test our entire system, we took it inside and tested it on the three tasks it is supposed to carry out. Namely, video streaming, obstacle detection, and monitoring. The encouraging part is it all worked perfectly.

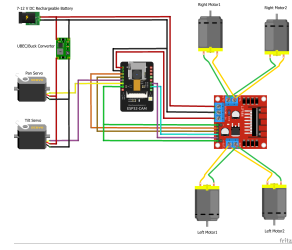


Fig. 1. System Architecture of the Surveillance Car

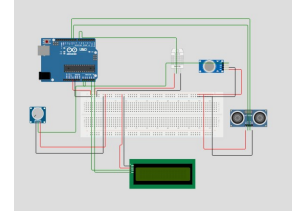


Fig. 2. Assembled Prototype of the Sensors

1) *Video Streaming:* The ESP32-CAM is the core module in this setup really worked very well. The live stream received through this component was clear and Almost instantaneous or with negligible delay times. This is very critical, since this is used to control the robot remotely without the control being laggy. The most critical Part here is the aspect of capability to rotate the camera. by 360° using a servo motor, which allowed us in order to capture all the surroundings without moving the entire car.

2) *Obstacle Detection and Response:* 2) It Sees and Stops itself, the used ultrasonic sensor worked perfectly well in detection of objects ahead of it. In an instance when it detected an approaching obstacle, It immediately triggered sending of the signal to the central controlling body, ESP32-CAM. The system not only detected the problem, but it acted. immediately acted on it. It quickly raised the signal to send the "STOP" signal to the wheels thereby stopping the device. This happened at the immediate moment to avoid any accident. In parallel with the triggering signal being forwarded to the operator in control from a distance.

3) *Environmental Monitoring:* 3) Environmental Monitoring: The team provided accurate data from different issues, such as temperature and air. quality. The data obtained is displayed on the small screen. of the gadget. More importantly, all this Real-Time Data, On-Screen and Online The two environmental sensors The experiment involved members of the data gathering team. whose job is to monitor and follow up on data about the conditions inside, s data is wirelessly connected to our remote dash board Now, remotely, this dashboard allows us to track the live feed and observe current temperatures and air quality. Measurements side by side, all from another room. Table I shows sample readings.

C. Discussion

In the provided setup, two controllers were employed: in order to make the system efficient. The first controller is in charge of the polling mechanism for data acquisition. information from the sensors is accurate and continuous. The other The controller participates in the networking

TABLE I
SAMPLE ENVIRONMENTAL DATA READINGS

Parameter	Value	Unit
Temperature	28.5	°C
Humidity	65	%
Gas Level	150	PPM
Dust Density	40.2	µg/m ³

process. dealing with the more challenging task of stable live delivery. video streaming. There is no interaction between the two processes, and hence the system is capable of performing efficient data acquisition as well as remote control functions along with stable live video streaming

IV. CONCLUSION

A. Summary of Outcomes

What began as an idea on the whiteboard, developed through brainstorming and late night diner discussions has now been realized, the robotic guardian is finished and completely functional. This forms the successful culmination of our mission to develop an overall, intelligently interconnected surveillance car. We didn't just connect devices; we realized a platform where the virtual and real worlds are combined in perfect synergy. The finished prototype is the expression of our team's efforts—a roaming watchdog that offers a live, all-inclusive view of remote locations, sampling the very air that it pervades. Naturally it defends itself against any collision. The achievement of this task within the framework of a sound two-controller- IoT-based overall platform, as the main task of this assignment. The observation of the entire system working in synergy, as one whole reactive system, is the most validating experience we have obtained from our philosophy of design. This, in fact, goes beyond a combined sensor and motor system—that's only a proof-of-concept, prototype of an altogether simpler system of automatic observation

B. Technical Achievements and Lessons Learned: The Heart of the Endeavor

By establishing a steady, efficient serial A communication link was the nervous system of our project. We learned to appreciate the elegance of distributed processing making Arduino act like a very fastidious data gatherer, while the ESP32-CAM flexed its power as a network maestro and a visual storyteller. This division was not without challenges. We spent a lot of hours hunched over our workbench, oscilloscopes probing signal lines, Debugging why an ultrasonic sensor reading would occasionally glitch and go off without any due cause. Reliability in embedded systems are won through relentless attention to detail, from ensuring clean power delivery to writing fault tolerant code. Beyond just the code, we learned some deep lessons in collaboration and project management. We navigated the complexities of merging code from different team members, managing an inventory of shared parts, troubleshooting as unified team. The moment the car responded for the first time to a command from our web interface, turning smoothly as the smoother, the video streamed without lag was a jolt of pure exhilaration. That made every previous setback worthwhile. This project forged us not just as programmers and engineers, but as problem-solvers.

C. Social Relevance and Future Scope

The value of our project goes far beyond the laboratory. In an increasingly security-conscious world, Concerned about both environmental health and operational efficiency. We have showcased that sophisticated monitoring requires not be prohibitively expensive or complex. We envisage this low-cost, flexible platform being the silent guardian of small businesses, the pioneering explorer in hazardous. Environments unsafe for human entry and the engaging classroom instrument to inspire the next generation of innovators in classrooms and makerspaces. Its capability of a real-time situational overview would democratize security. For the communities and yield extremely valuable data in industrial settings. This is not the end of the project for us, but a start. For this is a sure foundation we are passing on, a platform with potential for expansion in the future, the road for those who wish to carry on, the future is bright and clear. This work includes:

- 1) GPS Integration for Spatial Awareness: Inclusion of a GPS module would impart the capability of the car to understand not only the environment but also where it is in that environment. Further, it would enable route logging, geofenced autonomous patrols, and creation of real-time environmental heatmaps that visually correlate sensor data with specific locations.
- 2) A Custom Mobile Application: More personal and accessible, the system would come with an enhanced mobile application on all platforms rather than merely having it on a web-based dashboard. Imagine being able to control the guardian from your smartphone, receiving push-notification alerts, and seeing sensor histories with one tap.
- 3) Advanced Autonomous Navigation Algorithms: Obstacle avoidance nowadays is a very significant reflex; the next step in evolution would be truly to avail the car of its intelligence. The implementation of path-planning algorithms like A* or potential fields would transform it from a reactive machine into an active explorer, capable of making its way through complex environments in order to complete missions independently of direct human guidance. Eventually, this is more than our surveillance car; it is a symbol of what small, dedicated teams can achieve. This represents the power of integrating simple systems into complex, meaningful solutions. We are really proud of what we have built and are excited to see where this guardian will go next.

V. APPENDIX



Fig. 3. survey 1



Fig. 4. survey 2

A. Servey 2 details

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As discussed in [1], surveillance cars using ESP32-CAM modules provide live video streaming and obstacle detection. Similarly, in [2], IoT-enabled robots are used for monitoring hazardous areas.